MAGNETIC FORCE-BASED MICROFLUIDIC CHIP FOR PLANT SEED LEVITATION TO SIMULATE MICROGRAVITY ENVIRONMENT

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ABSTRACT

For plants on Earth, gravitropism is essential for seedling establishment and plant growth. As an integrated part of plant development, gravitropic response is mainly mediated by changes of photohormone auxin. Study on auxin responses in microgravity condition is critical for illustrating how plants perceive environmental gravity conditions and then initiating auxin signaling pathway, ultimately, helping to reveal how plants response to gravity stimulus. Nevertheless, key barriers to answering these questions are the lack of a tool that can provide simulated microgravity condition. In this work, we have developed a microfluidic negative-magnetophoresis (MNM) platform to levitate Arabidopsis seeds in an equilibrium plane where magnetic force and buoyancy force are equal and opposite in directions. With the advantage of the MNM platform to simulate a microgravity environment, it's found that auxin response is obviously decreased after seed levitation during early seed germination. This can enable the study of the physiological process and molecular mechanism of auxin in a simulated microgravity condition and benefit further applications for optimizing plant growth in space.

KEYWORDS

Magnetic force, simulated microgravity, levitation, Arabidopsis seeds, auxin.

INTRODUCTION

Gravity is unique and constant factor on Earth throughout the entire life cycle. Plants are sessile and well adapted to this 1g level, which is used by plants to evolve gravitropsim in establishing development pattern and optimizing the utilization of light, water and mineral salts from 500 million years ago [1]. Consequently, the alteration of this parameter profoundly influences physiological processes and initiates adaptive responses. As a cardinal regulator of all aspects of plant growth and development [2], auxin has involved in responses to gravity stimulus through redistribution of auxin gradients [3]. It's important to study the role of auxin during plant development in microgravity condition, which is essential for plant space biology.

Normally, the way to encounter microgravity environments is merely in space or during free fall. The International Space Station (ISS) or Space Shuttle provides short lasting condition in the cost of long-duration spatial missions [4]. Free fall tower is comparably lack of providing continues and long-term microgravity. More importantly, either method is applicable for routine laboratory research. Therefore, simulating microgravity devices have been developed. Random Positioning Machine (RPM) can mimic microgravity through rotary system, such as 1D-clinostat, 2D-clinostat and 3D-clinostat [5]. Although RPM generate comparable effects of microgravity when the direction changing is fast enough, the limitation is to monitor several individual samples simultaneously.

The other technology to mimic microgravity is magnetic levitation technique, which can be applied via positive or negative magnetophoresis. However, positive magnetophoresis levitates the subject of interest based on modification of surfaces, thus the internal structures are free of those forces [6]. In contrast, negative magnetophoresis-mediate levitation can exactly mimic microgravity through counteracting gravitational force in a manner of free labeling. In this technique, objects are triggered towards regions of low magnetic field in magnetic field gradients, resulting in stable magnetic levitation and the simulation of microgravity condition [7]. To reduce the magnitude of magnetic fields, simulated microgravity can also be compromised by increasing magnetic susceptibility of ferrofluids or paramagnetic medium. In the label-free levitation, gadolinium (Gd^{3+}) based solution has been gradually utilized, due to good biocompatibility, high efficiency and comparable inexpensive [8-10].

In this study, we used MNM in a microfluidic chip to establish a magnetic levitation system for *in situ* plant seed culture in simulated microgravity. The development of such platform has led not only to achieving better understanding of the microgravity phenomenon but also of its application to diverse biological samples.

EXPERIMENTAL

Levitation Device Fabrication

The microfluidic device was realized by plasma bonding between a polydimethylsiloxane (PDMS) with microfluidic structures and a glass substrate. The PDMS layer was fabricated by soft lithography. In brief, a 180 µm thick layer of SU8 2100 (19060515, MicroChem Corp.) was spin-coated onto a silicon wafer and soft baked at 95 ºC for 35 minutes. The wafer was then exposed to the photomask containing the chip design at 280 mJ/cm^2 using a EVG mask aligner (610TB, EV Group GmbH). After the post exposure bake at 95 ºC for 13 minutes, the wafer can be placed in a dish of SU-8 developer for approximately 12 minutes, gently agitated until the exposed features remain. The wafer was rinsed with clean IPA and then dried with N_2 gas. The master was then used to mold PDMS (Sylgard 184, Dow Corning Corp.). PDMS was cast at 1:10 ratio (curing agent/base elastomer), degassed under vacuum, and cured in an oven for 2 hours at 85 ºC. Access holes were punched on the PDMS layer, which was then completely cleaned for oxygen plasma bonding.

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Simulation

Governing equations

- Magnetic force

The magnetic force F_m is given by Eq. (1):

$$F_m = \frac{V_s(\chi_s - \chi_m)}{\mu_0}(B \cdot \nabla)B \quad (1)$$

where μ_0 is the permeability of free space, V_s is the volume of seed, χ_s and χ_m are the magnetic susceptibility of the seed and Gd^{3+} solution (381667, Sigma), respectively. The relationship between magnetic susceptibility and paramagnetic salt concentration is given by Eq. (2) [11]:

$$\chi_m = (0.35 \times C_{Gd} - 9.04) \times 10^{-6} \quad (2)$$

where C_{Gd} is the concentration of Gd^{3+} solution (mM).

- Hydrodynamic drag force

The hydrodynamic drag force F_d is given by Eq. (3):

$$F_d = 6\pi\eta_s r\Delta v f_D \quad (3)$$

where r is the radius of particle (here the semi-major axis a of the ellipsoid is instead of r), Δv is the velocity difference between the seed and the Gd^{3+} solution, η_s is the viscosity of the Gd^{3+} solution, f_D is the drag coefficient of the particle and incorporates the influence of a solid wall in the vicinity of the moving seed, given by Eq. (4):

$$f_D = \left[1 - \frac{9}{16}\left(\frac{r}{r+z}\right) + \frac{1}{8}\left(\frac{r}{r+z}\right)^3 - \frac{45}{256}\left(\frac{r}{r+z}\right)^4 - \frac{1}{16}\left(\frac{r}{r+z}\right)^5\right]^{-1} \quad (4)$$

z is the distance of the seed to the solid wall.

- Viscous resistance

The gravity F_g is given by Eq. (5):

$$F_g = m_s g \quad (5)$$

where m_s is the weight of a single seed and g is the gravitational constant.

- Buoyancy

The buoyancy F_b is given by Eq. (6):

$$F_b = \rho_m g V_s \quad (6)$$

where ρ_m is the density of Gd^{3+} solution.

Based on the governing equations above, we used COMSOL software to simulate the movement of the seed in Gd^{3+} solution.

Plant Growth and Seed Harvest

The Arabidopsis wild-type and transgenic plants used in this study were in a Col-0 (Columbia) background, ordered from Nottingham Arabidopsis Stock Centre (NASC). All plants were grown in a controlled LED-light grow chamber (MGC-100HP-2L, Shanghai Yiheng Technical Co., Ltd.) at 20-22 ºC under cool-white fluorescent light (80-100 μmol/m^{2}s) in a long-day photoperiod (16 h light/8 h dark). Arabidopsis seeds were harvested only after the siliques have completely browned.

Seed Stratification and Germination

Surface of Arabidopsis seeds were sterilized using 50% (v/v) bleach solution in tubes [12]. After sterilization, seeds were suspended in autoclaved distilled water and then kept at 4 ºC in refrigerator for 2 days to break seed dormancy. After refrigerated stratification, seeds were sowed in cylindrical channels (1000 μm in diameter) on chip and cultured and germinated with half-strength Murashige and Skoog (1/2 MS) medium for 1 day.

Cryostat Slicing and fluorescent labeling

The seeds were embedded in Optimal Cutting Temperature compound (OCT, 23-730-571, Fisher HealthCare) in a container. Then, OCT-embedded seeds were snap frozen and stored in the container at -80 ℃ in refrigerator for at least 2 hours. Sliced sections (25 μm) were cut from one side of the seeds to the opposite side continuously using a cryostat (CM1950, Leica). From approximately 20 slides per seed, each slide was placed on a glass slide and dried at room temperature. These slides were then washed with phosphate buffered saline (PBS, pH7.4) buffer to remove OCT and fixed with paraformaldehyde (PFA, 158127, Sigma) for 10 minutes. After washing in PBS buffer again, the slices were mounted with DAPI solution (10236276001, sigma) and sealed with cover glasses.

RESULTS

The Levitation Microfluidic Device

The microfluidic device for seed levitation fabricated by soft lithography is shown in Figure 1A. The chip is consisted of 5 straight microchannels which are arranged in parallel, each microchannel has 2 cultivation reservoirs to independently control the levitation process of a single seed in a reservoir. Thus, a single chip enables simultaneous operation of 10 seeds at the same time (see Figure 1B). Below the glass substrate, there are 2 NdFeB permanent magnets (N52, 1" × 1/4" × 1/4") in opposite polarities providing magnetic force (see Figure 1C). The distance between two permanent magnets is 1000 μm, and the thickness of glass substrate is 200 μm.

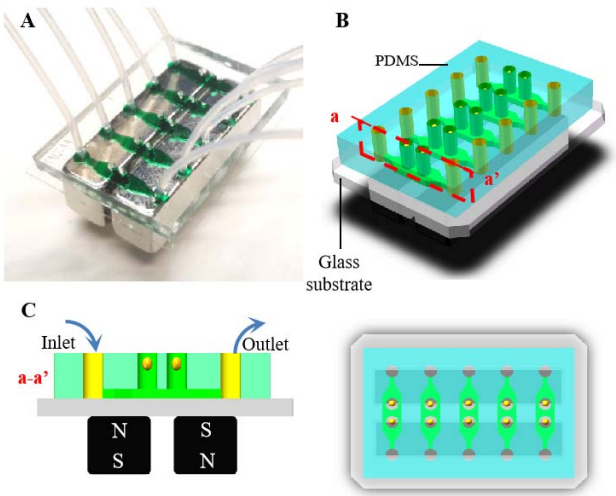

Figure 1. Chip design and experiment procedures. (A) Photo of the microfluidic chip for seed levitation. (B) Design of the chip with 5 micro-channels for seed cultivation, each channel (10 mm in length, 2 mm in width and 100 μm in height) has two cultivation reserviors (1000 μm in diameter). (C) Layout of the microdevice showing the position of the magnets and the plant seeds.

In order to cultivate the seeds in the best suspension state, we established a simulation model of a seed in the magnetic solution. COMSOL software was used to qualitatively analyze the force and suspension trajectory

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under the action of the permanent magnet. Accordingly, the position of the cultivation reserviors was optimized (see Figure 2A). Here the seed is simplified to an ellipsoidal (a = 0.25 mm, b = 0.175 mm, c = 0.175 mm), the initial vertical height of its center is 0.2 mm (Y = 0.2 mm), and the initial horizontal position is changed from 0 to 2.5 mm. The seeds in the Gd^{3+} solution are subjected to the magnetic force F_m, hydrodynamic drag force F_d, gravity F_g and buoyancy F_b (see Figure 2B), the magnetic force is the main power to provide suspension, which is given by

$$F_m = \frac{V\Delta\chi}{\mu_0}(B \cdot \nabla)B \qquad (7)$$

where μ_0 is the permeability of free space, V is the volume of seed, $\Delta\chi$ the difference in magnetic susceptibility between the seed and the Gd^{3+} solution. The main factors that determine the magnitude of the magnetic force are the magnetic field strength and the magnetic field gradient. The simulation results show that near the edges of the two magnets, both the magnetic field strength and the magnetic field gradient reach the maximum values, so the magnetic force near the magnet edge (X = ± 0.5 mm) is greater than that in other positions, and the height of the seeds in the Y direction is also higher, however, the magnetic force and the height of the suspension are very small in the position away from the edge of the magnets (X = ± 2.5 mm), as shown in Figure 2C, since the magnetic field is symmetric about the Y axis, we only show the results of the positive X axis here. The above simulation results show that the area above the two magnet edges is the most suitable for seed suspension, so the cultivation reservior is designed in the area above the magnet edge (see Figure 1).

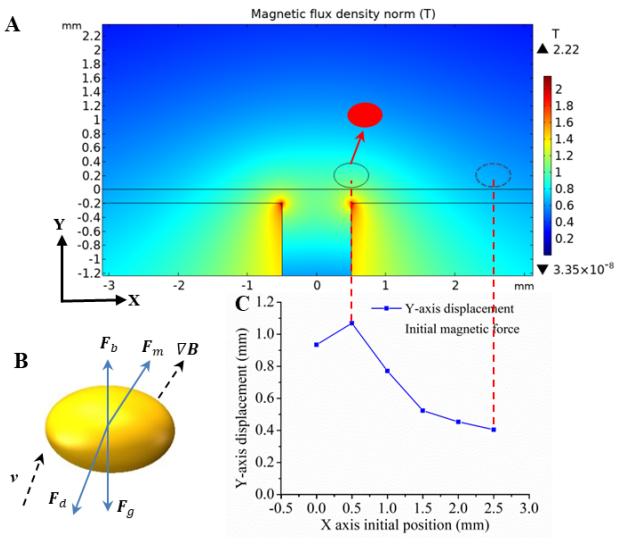

Figure 2. System optimization for seed levitation using COMSOL Multiphysics. (B) The force analysis of seed in Gd^{3+} solution. (C) The simulation results of initial magnetic forces and Y-axis displacements of seeds in different X-axis positions.

Before initiation of Gd^{3+}-promoted seed levitation, the toxicity of Gd^{3+} solution on seed germination was test. It's been observed that seeds immersed in Gd^{3+} solution for 30 minutes represented limited difference from those untreated seeds during subsequent germination process (see Figure 3A). Thus, 200 mM Gd^{3+} is a safe condition for plant seeds during short period of stimulated levitation. To shape the ellipsoid-like seeds, the parameters a, b and c (the lengths along x, y, z axis) are measured and calculated according to microscope images. After statistical analysis, the lengths of a, b and c are 578.0 ± 60.6 μm, 406.1 ± 30.5 μm, 335.6 ± 40.7 μm, independently. To ensure Gd^{3+}-promoted seed levitation in solution, the half of the longest axis of seeds (> 300 μm) is the levitated threshold of simulated microgravity. According to this standard, it's been observed that seed floating can be guaranteed at the concentration of 200 mM, due to the levitation distance is over 400 μm (see Figure 3B). Thus, 200 mM Gd^{3+} is a safe and sufficient condition to create a simulated microgravity niche for Arabidopsis seeds on the chip.

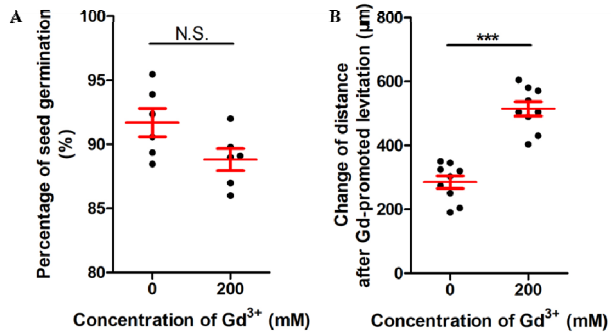

Figure 3. Gd^{3+}-promoted magnetic levitation of Arabidopsis seeds in situ on chip. (A) Quantification of percentage of seed germination after treatment of the indicated Gd^{3+} solutions for 30 minutes. Data are mean values ± SE, with six biological repeats for each bunch of Arabidopsis seeds (n=30). (B) Histograms of levitation height change of seeds at indicated of Gd^{3+} solutions. Data are mean values ± SE. Each dot represented one of ten seeds in the chip device.

Auxin is an essential phytohormone contributes to all aspects of the plant development. As the indicator of auxin response, transgenic plant DR5::GFP was chosen (see Figure 4). Before the examination of auxin response to stimulated microgravity, auxin responses of embryo to Gd^{3+} solution were examined. It's been observed that short-term treatment of Gd^{3+} hardly shows effect on auxin signal during early germination (see Figure 4B). Through application of magnets under Gd^{3+} solution, seeds were levitated and then kept for 30 minutes. It's been observed that auxin response is dramatically repressed after seed levitation (see Figure 4C). Auxin responses are the downstream reflection of spatiotemporal distribution of auxin, depending on not only its metabolism but also its directional cell-to-cell transportation [2]. In future, the studies of magnetic levitation on auxin synthesis and transportation signaling pathway will help to illustrate underlined mechanism of auxin response to microgravity.

CONCLUSIONS

Auxin has been established as one of critical factors in mediating plant gravitropic responses. However, due to difficulties to generate microgravity condition on Earth, the widely used microgravity simulator in lab cannot

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provide a spatio-temporal observation of plant on cellular level. We designed and fabricated a microfluidic chip for long-term culturing of Arabidopsis seeds. With application of Gd^{3+}-protomted negative magnetoresistance, plant seeds were reservedly levitated on chip. In the meantime, auxin response of plant embryo was found with an obvious decrease after simulated microgravity. In future, the mechanism of auxin response to microgravity in plant can be further studied. Furthermore, due to convenience and usefulness of this approach, the MNM platform will be widely applied to variety of biological samples.

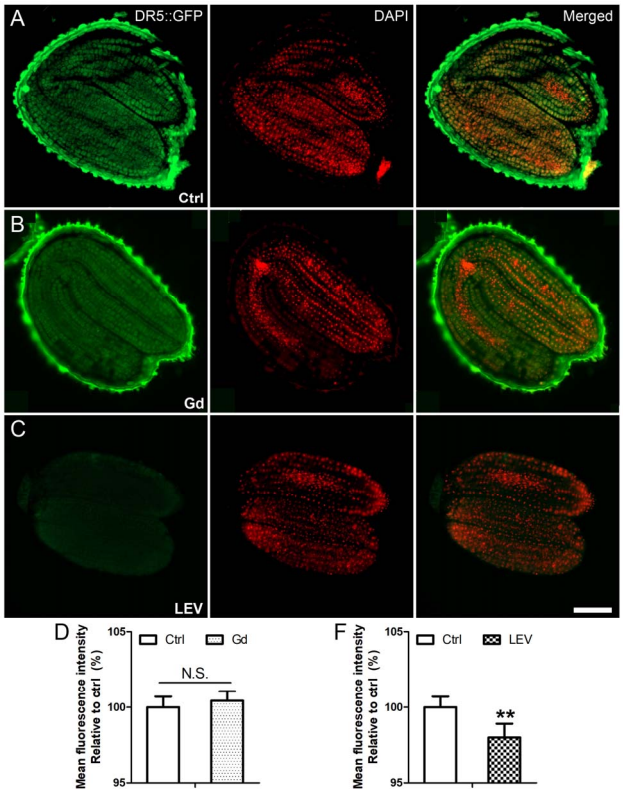

*Figure 4. Magnetic levitation represses auxin response during Arabidopsis seed germination. (A-C) Expression of DR5::GFP in embryo sections of Arabidopsis seeds without (A), with Gd^{3+} (B) and Gd^{3+}-mediated magnetic levitation (C). The red is fluorescence from 4′,6-diamidino-2-phenylindole (DAPI) dye, the green is GFP and the oval outline in strong green is auto-fluorescence from seed coats. Scale bar= 100 μm. (D-F) Quantification of relative expression of DR5::GFP in embryo sections of Arabidopsis seeds without, with Gd^{3+} (D) and Gd^{3+}-mediated LEV (F). Error bars represent standard error (SE) of the mean (n>30) with three biological repeats. Asterisks indicate significant differences (Student's t-test, **, P≤0.01).*

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